Then

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

$$= \left|\frac{\pm 2\sqrt{h^2 - ab}}{a + b}\right|$$

Let us take 'θ' as acute angle

$$\therefore \quad \theta = \tan^{-1}\left[\left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|\right]$$

NOTE

(i) If θ is the angle between the pair of straight lines

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$,

then $\theta = \tan^{-1}\left[\left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|\right]$

(ii) If the straight lines are parallel, then $h^2 = ab$.

(iii) If the straight lines are perpendicular, then $a + b = 0$

i.e., *coefficient of* x^2 + *coefficient of* $y^2 = 0$

3.3.4 The condition for general second degree equation to represent the pair of straight lines

The condition for a general second degree equation in x, y namely

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

to represent a pair of straight lines is

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0.$$

NOTE

The condition in determinant form is

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

Example 3.10

Find the combined equation of the given straight lines whose separate equations are $2x + y - 1 = 0$ and $x + 2y - 5 = 0$.

Solution

The combined equation of the given straight lines is

$$(2x + y - 1)(x + 2y - 5) = 0$$

i.e., $2x^2 + xy - x + 4xy + 2y^2 - 2y - 10x - 5y + 5 = 0$

i.e., $2x^2 + 5xy + 2y^2 - 11x - 7y + 5 = 0$

Example 3.11

Show that the equation $2x^2 + 5xy + 3y^2 + 6x + 7y + 4 = 0$ represents a pair of straight lines. Also find the angle between them.

Solution

Compare the equation

$2x^2 + 5xy + 3y^2 + 6x + 7y + 4 = 0$

with

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$,

we get

$a = 2, b = 3, h = \frac{5}{2}, g = 3, f = \frac{7}{2}$ and $c = 4$

Condition for the given equation to represent a pair of straight lines is

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0.$$

$$abc + 2fgh - af^2 - bg^2 - ch^2 =$$

$$24 + \frac{105}{2} - \frac{49}{2} - 27 - 25 = 0$$

Hence the given equation represents a pair of straight lines.

Let θ be the angle between the lines.

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